Governance & Sybil boundaries

An analytical companion to the DAO project

STATUS: Project-hosted technical note. The linked SSRN preprint is a separate public research artifact. This document is not a Zenodo archive or a peer-reviewed article.

1. Public paper and separate model

The Arbitrum preprint reports outcome changes in 4 of 15 selected proposals under square-root reweighting. See SSRN abstract 7103878. The identity-splitting preview below is a separate fixed-budget calculation, not a replication of the paper's ballot-level analysis.

2. Derive the equal-split amplification

Let T be a fixed token budget and W = sqrt(T) its unsplit square-root weight. Splitting equally across k identities gives total weight k sqrt(T/k) = sqrt(k) W. The additional weight is (sqrt(k) - 1) W. Equal splitting maximizes total square-root weight for fixed k and T, assuming fractional holdings and no identity costs.

If G is the existing winning-side minus losing-side score gap, with the attacker's unsplit weight already included on the losing side, strict reversal requires (sqrt(k) - 1) W > G. For W > 0 and G >= 0, the smallest integer k is floor((1 + G/W)^2) + 1. Equality creates a tie, not a strict reversal.

3. Algebraic checks of published summaries

Summary label	W	G	Strict k minimum
Grants Milestone 1	4,272.48	3,992.73	4
ACryptoS LTIPP	4,252.89	16,856.83	25
Arbitrum Coalition	5,192.48	73,484.43	230
Security Enhancement	3,834.53	110,816.18	894

Provenance & limitations

4. What was checked

The four W/G pairs are hard-coded in dao-sybil-resilience/run_sybil_pipeline.py. The site recomputes each integer boundary and verifies that the additional weight is above G at k and no greater than G at k - 1. evidence/sybil-boundaries.json records the values, derivation, source URL and source-file SHA-256.

The companion arbitrum-quadratic-voting-simulation repository contains Snapshot GraphQL collection, vote reweighting, concentration metrics and reconstruction checks. Proposal metadata is committed. A full frozen raw-ballot export is not committed in the inspected repositories, so the four W/G summaries cannot be independently reconstructed from those repositories alone.

5. Interpretation boundaries

A wallet is not necessarily a person. Splitting can involve transaction fees, identity requirements, minimum balances, delegation constraints and limits on eligible balances. The preview assumes these away, holds the opposing tally fixed and permits equal splitting of a fixed budget. It does not estimate the cost or feasibility of an actual attack.

The browser preview and the phase-boundary checks are analytical calculations. They do not establish that a specific minority token share can overturn a real proposal through three wallets, nor that changing a vote-weight rule leaves voter behavior unchanged.

6. Reproduce and extend

```
python -m pytest tests/test_benchmark_evidence.py -q
python scripts/build_portfolio.py
```

Next steps: archive the exact vote records with proposal IDs and timestamps; verify page boundaries and reconstruction totals; publish the frozen outputs; and add identity-cost, eligibility and integer-token constraints before interpreting real-world exploitability.

Source links

github.com/thetarocks/dao-sybil-resilience
github.com/thetarocks/arbitrum-quadratic-voting-simulation
papers.ssrn.com - abstract 7103878